

Adrian Legaspi

Backend Engineer

Monterrey, Mexico

Profile

Senior Software Engineer with 9+ years building backend services and REST APIs in Node.js, TypeScript and Python, across enterprise security, fintech and AI products.

Designs ontologies and semantic data models in OWL, builds API-first and event-driven services on AWS, and ships LLM applications with RAG retrieval, vector search and agent orchestration. Professional-level English for technical and conversational communication.

Experience

Software Engineer, JointechAI – Tijuana, Mexico

Aug 2022 – present

- **Viasat Internal Security Application (via Borderux)**: Designed and maintained the platform ontology and semantic data model, formalized as an OWL class and property hierarchy: products, the findings raised against them, and the assets that discover those findings. Government-contract security program; specifics under NDA.
- Maintained and evolved that taxonomy as the domain grew, keeping products, findings and the assets that discover them queryable as one connected model instead of per-source silos.
- Integrated heterogeneous scanner and asset sources into that model through event-driven Lambda handlers, S3 event triggers, and inbound webhooks, with Redis caching on hot read paths and CloudWatch logs and alarms for operational visibility.
- Delivered Python AWS Lambda functions and REST API services behind API Gateway, each deployed and versioned independently, embedded in a backend-heavy team of 15.
- Built the centralized React and TypeScript dashboard that aggregates security findings for 4,000+ internal users, as the sole frontend engineer on the team.
- Worked across the platform AWS environment, including S3, EC2, CloudWatch and Lightsail, with services running in Docker containers and releases shipped through GitHub Actions.
- **Agentic HR ERP (Jointech AI)**: Tech Lead for a team of 5 engineers on the internal HR platform managing 30 employees and analyzing prospective roles and candidates for hiring.
- Built its Node.js, Express and TypeScript backend: REST API design, service integration, and the data model behind it, containerized with Docker across local and deployed environments.
- Implemented the RAG architecture covering document ingestion, chunking, embeddings and Chroma vector search, delivering grounded semantic search across heterogeneous HR and candidate documents.
- Developed agent orchestration and prompt-engineering workflows over Model Context Protocol, wiring retrieval and structured tools into LLM calls, plus model fine-tuning workflows. Built the React client on top of those services.

Software Engineer, Arte Digital – Tijuana, Mexico

Jan 2020 – Aug 2022

- **Captira**: Designed and built secure Laravel REST APIs serving 3,000+ legal professionals, with the relational data model behind them.
- Built complex Angular data dashboards on those APIs, covering case, document and reporting workflows.
- **Crediverso**: Architected and shipped full-stack fintech features for a platform serving 10,000 users, on React and a headless WordPress backend.
- Established the CI/CD pipelines with Codeship, later migrating them to AWS CodePipeline, cutting deployment time for the team.

Web Designer & Developer, LuballSoftware (Freelance) – Tijuana, Mexico

Mar 2016 – Jan 2020

- Developed and managed FireApp, a digital logging application for the Tijuana Fire Department used by 180+ field firefighters and paramedics across 18 stations for pre-hospital incident reports.
- Delivered 25+ web solutions for local businesses over 4 years, ranging from custom e-commerce platforms to landing pages.

Education

Universidad Tecnológica de Tijuana, BS in Tecnologías de la Información – Tijuana, Mexico

May 2015

Skills

Ontologies & Semantic Data: OWL ontologies, semantic data models, taxonomies, class and property hierarchies, domain modeling

Backend & API Engineering: Node.js, Express, TypeScript, Python, REST API design, API-first contracts, AWS Lambda, API Gateway, PostgreSQL, Redis, Firebase, Laravel (PHP)

Architecture & Integration: Serverless microservices, event-driven handlers (S3 events, webhooks), asynchronous workflows, service-to-service integration, data integration patterns, Model Context Protocol

Generative AI & Semantic Search: RAG architectures, Chroma vector search, embeddings, semantic retrieval, agent orchestration, prompt engineering, fine-tuning, Gemini APIs

Cloud & DevOps: AWS (Lambda, API Gateway, S3, EC2, CloudWatch, Lightsail), Docker, CI/CD (GitHub Actions, AWS CodePipeline, Codeship), software architecture

Frontend: React, TypeScript, Angular, SCSS

Languages

English: Professional proficiency

Spanish: Native

Side projects

Shroom Merge ([Play Store](#), [App Store](#))

July 2026

- Independently developed and released Shroom Merge (Kinoko Merge on iOS), a mobile merge game, taking the project from concept to Play Store and App Store launch.